MA265 CONDENSED NOTES

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1. Classification of PDE's

We can split the classification of PDE's into two types. There are criteria that apply to all PDE's, and there are criteria which apply **only** to second-order linear PDE's.

All PDE's

- Order of a PDE.
- Linearity (in the functions and its partial derivatives).
- Homogeneity.

In general, to assess these things, we can rewrite our PDE in the form $\mathcal{L}(u) = r(x)$, where $\mathcal{L}$ is the differentiation operator (a function of u and all its partial derivatives) and r is 'everything else'. Then we assess the above criteria by looking at $\mathcal{L}$ in the case of order and linearity, and by looking at r in the case of homogeneity (homogeneous if $r \equiv 0$.)

Second-Order Linear PDE's

In order to assess these things, we first rewrite our PDE in the following form:

$$A\partial_{xx}u + B\partial_{xt}u + C\partial_{tt} + \cdots = 0$$

We call the above the *leading part* of the PDE. We define the *discriminant* as usual: $D = B^2 - 4AC$. Then we have the following classification.

- $D > 0$. Hyperbolic. The wave equation is an example.
- $D = 0$. Parabolic. The heat equation is an example.
- $D < 0$. Elliptic. The laplace equation is an example.

The above is also the order in which we cover these PDE's in the course. Remember by noticing that the discriminant is decreasing as we progress throughout the course.

Inital and Boundary Conditions

We require these for well-posedness, (in particular uniqueness). There are two main distinctions to be made.

- Unbounded domain Ω, initial values given $\implies$ IVP/Cauchy Problem.
- Bounded domain Ω, initial values and boundary conditions given, BVP (Boundary Value Problem).

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There are four main types of Boundary Conditions, all used for modelling different things.

- Dirichlet Boundary Conditions
  - $u(x, t) = u_D(x)$ for all $x \in \partial\Omega$.
  - If $u_D(x) \equiv 0$, we say that this is a homogeneous Dirichlet BC.
  - Uses: Fixing the boundary, e.g. population of a species.
- Neumann Boundary Conditions
  - $\partial_n u(x, t) = u_N(x)$ for all $x \in \partial\Omega$.
  - ∂_n denotes the parital derivative in the normal direction to the boundary.
  - $u_N(x) \equiv 0 \implies$ homogeneous Neumann BC.
  - Uses: Modelling how change occurs at the boundary.
  - Homogeneous Neumann BC of temp of a rod models perfect insulation.
- Mixed/Robin Boundary Conditions
  - $\partial_n u(x, t) = \alpha(u(x, t) - u_R(x))$ for all $x \in \partial\Omega$.
  - Not entirely sure what the point of this is quite yet.
- Periodic Boundary Conditions
  - Suppose we are solving a PDE on a circle. One way in which we can do this is by solving on $[0, 1]$ and adding the condition that $u(1, t) = u(0, t)$. This is a periodic boundary condition.

Well-Posedness

Stability, Uniqueness, Existence. Give examples of where each fails when this comes up in future reading of the notes.

2. The Transport Equation.

Derivation

The transport equation arises from the simple conservation of mass law, which can be stated as follows: 'rate of change of total mass = inflow - outflow'. Consider a tube filled with some quantity, of constant cross-sectional density, and cross-section area A. The tube goes from $x = a$ to $x = b$. Denote by $u(x, t)$ the density of the cross-section at x at time t. Denote by $q(x, t)$ the flux (inflow/outflow) of the quantity. Then, we may formulate the law of conservation of mass mathematically as follows:

$$\frac{d}{dt}\int_a^b Au(x, t)dx = Aq(a, t) - Aq(b, t)$$

Appling FTC (assuming everything is nice enough), and pulling the derivative inside gives

$$A \int_a^b \partial_t u(x,t) + \partial_x q(x,t) dx = 0$$

Since a, b are chosen arbitrarily, and since the integrand is continuous, it follows that the integrand must be identically zero. Thus, we get the PDE

$$\partial_t u(x,t) + \partial_x q(x,t) = 0 \tag{1}$$

This isn't the transport equation just yet, we get there by using various 'constitutive laws' - those which relate the density and the flux. A simple example is to say that the density and the flux are proportional. Mathetmatically this translates to $cu(x,t) = q(x,t)$ for some constant c. In this case (1) becomes the *linear advection equation.*

$$\partial_t u(x,t) + c\partial_x u(x,t) = 0 \tag{2}$$

We can generalise linear advection slightly to get the *transport equation.*

$$\partial_t u(x,t) + v(x,t)\partial_x u(x,t) = 0 \tag{3}$$

The Method of Characteristics

First, we begin by fully stating the problem. Finding a solution to the *transport equation* is the problem of finding a function u that satisfies

$$\begin{aligned}
&\partial_t u(x,t) + v(x,t)\partial_x u(x,t) = 0, \ \forall (x,t) \in \mathbb{R} \times [0,T] \\
&u(x_0, 0) = \phi(x_0) \ \forall x_0 \in \mathbb{R}
\end{aligned}$$

Where v and ϕ are some given, 'sufficiently nice' functions.

The idea behind the method of characteristics is to find curves in the $x - t$ plane where u is constant. Then, we may travel along them until we reach the x-axis, at which the function value is known. These are called *characteristic curves*, and they are defined as follows:

$$\begin{aligned}
&\xi'(t) = v(\xi(t), t) \\
&\xi(0) = x_0
\end{aligned}$$

One can verify that $u(\xi(t), t)$ is a constant function of t, simply by differentiating with respect to t. So, we simply solve, substitute in appropriately, and we are done.

Two Warnings

The method of characteristics can go wrong in some ways. One way, is that there may be places where characteristic curves overlap. In which case, which one are we supposed to travel across to get to the x-axis? Another issue may arise when the characteristic curves don't cover the whole $x - t$ plane. In which case, this method cannot be used to determine the value of u at that point, since there is no curve to 'travel across' to get to the x-axis.

3. The Wave Equation

The wave equation in full generality is as follows:

$$\partial_{tt}u(x,t) = c^2\partial_{xx}u(x,t) \text{ for some } c \in \mathbb{R} \tag{4}$$

The analysis of the wave equation is very different depending on the domain on which we are considering it. One thing that we can do in either case, is to notice that the wave equation, in some sense, is two instances of the linear advection equation with opposite velocities. Interpreting ∂ as an operator, and assuming that u is C^2, we see that

$$(\partial_t + c\partial_x)(\partial_t - c\partial_x)u(x,t) = \partial_{tt}u(x,t) - c^2\partial_{xx}u(x,t)$$

Moreover, recalling that the solution to the linear advection equation with initial condition $u_0(x_0, 0) = u_0(x_0)$ has solutions $u_0(x - ct)$, we have a way of starting to analyse the wave equation. The coordinates $x \pm ct$ are called *characteristic coordinates.*

Analysis on the Real Line (The Cauchy Problem)

Before we do anything, we first fully state the problem. We are trying to find a function $u : \mathbb{R} \times (0,\infty) \to \mathbb{R}$ such that

$$\partial_{tt}u(x,t) = c^2\partial_{xx}u(x,t)\ \forall (x,t) \in \mathbb{R} \times (0,\infty) \tag{5}$$

$$u(x,0) = \phi(x)\ \forall x \in \mathbb{R} \text{ (IC1)} \tag{6}$$

$$\partial_t u(x,0) = V(x)\ \forall x \in \mathbb{R} \text{ (IC2)} \tag{7}$$

This is the IVP/Cauchy Problem. We are working on an unbounded domain, so we have no need to state boundary conditions. To find a solution, we first assume that one exists. Set $\xi = x + ct$, $\eta = x - ct$. Then, we write $v(\xi, \eta) = u(x,t)$ for the solution in the characteristic coordinates. Differentiating appropriately, and then integrating and some other stuff, we eventually conclude that

$$u(x,t) = f(x + ct) + g(x - ct)$$

for some unknown functions $f, g : \mathbb{R} \to \mathbb{R}$. These can be determined using the intitial conditions. After the calculation is said and done, we get the following:

$$u(x,t) = \frac{1}{2}(\phi(x+ct) + \phi(x-ct)) + \frac{1}{2c}\int_{x-ct}^{x+ct} V(r)dr \tag{8}$$

The above is called *D'Alembert's formula.* It remains to show that this is actually a solution, and that it is unique. Assuming that ϕ is C^2 and V is C^1, it follows from Leibniz' integral rule and the FTC (see Appendix) that this is a solution. To show existence, we use the *energy* associated to the equation (we actually only do this in the case of the BVP, presumably we can assume uniqueness?).

Analysis of the BVP - Dirichlet BC

As before, we begin by giving a full statement of the BVP with homogeneous dirichlet BC. Firstly, we can make a change of variable so that $c = 1$, $L = \pi$. Then, we are trying to find a function $u : (0,\pi) \times (0,\infty)$ such that

$$\begin{aligned}
&\partial_{tt} u(x,t) = \partial_{xx} u(x,t) \;\text{ for all } (x,t) \in (0,\pi) \times (0,\infty) \\
&u(x,0) = \phi(x) \;\text{ for all } x \in (0,\pi) \\
&\partial_t u(x,0) = V(x) \;\text{ for all } x \in (0,\pi) \\
&u(0,t) = u(\pi,t) = 0 \;\text{ for all } t \in (0,\infty)
\end{aligned}$$

We 'solve' this by seperating variables (then verifying that our solution is indeed a solution, then showing uniqueness using energy methods). Assume the solution is of the form $u(x,t) = X(x)T(t)$. Then, using the statement of the wave equation, we have that

$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{T(t)} = \lambda \in \mathbb{R}$$

This is constant because these fractions are functions of different variables yet equal. This gives us the following system of differential equations:

$$\begin{aligned}
X''(x) - \lambda X(x) = 0 \\
T''(t) - \lambda T(t) = 0
\end{aligned}$$

We now investigate what happens for different λ. The TL;DR is that for $\lambda > 0$, we get a contradiction, (integrate the above, multiply by $-X(x)$ and use integration by parts) $\lambda = 0$ implies that only the constant 0 function is a solution. For $\lambda < 0$, set $\lambda = -\beta^2$. Then, we get a solution for $X(x)$ of the form

$$X(x) = C\cos(\beta x) + D\sin(\beta x)$$

Moreover, $X(0) = 0 = X(\pi)$. The condition $X(0) = 0$ immediately implies that $C = 0$, and the condition $X(\pi) = 0$ implies that $\sin(\beta\pi) = 0$, thus $\beta \cdot \pi = \pi \cdot j$ for some $j \in \mathbb{Z}$. Thus, we have a solution for each j, and we denote the associated constant by D_j.

Solution the equation for $T(t)$ now, we see that

$$T(t) = A\cos(\beta t) + B\sin(\beta t)$$

But, from our remarks about X, we see that we get a solution every j,

$$T_j(t) = A_j\cos(jt) + B_j\sin(jt)$$

Thus, we get solutions for each j,

$$u_j(x,t) = (A_j\cos(jt) + B_j\sin(jt))\sin(jx)$$

And as the sum of solutions to linear PDE's is a solution, we get the general solution:

$$u(x,t) = \sum_{j\in\mathbb{Z}} (A_j\cos(jt) + B_j\sin(jt))\sin(jx)$$

This is worth memorising because the derivation is slightly annoying.

The BVP with homogeneous Neumann BC

The exact same procedure as above will yield the following general solution to the wave equation:

$$u(x,t) = c_0 + c_1 t + \sum_{j\in\mathbb{N}} (A_j\cos(jt) + B_j\sin(jt))\cos(jx)$$

Energy Methods

The *energy* associated to the wave equation is the following function:

$$E(t) = \frac{1}{2}\int_0^\pi (\partial_t(u))^2 + (\partial_x(u))^2 dt$$

The main result associated to energy is that if u is a solution to the wave equation, then $E(t)$ is constant.

4. Fourier Series

The above seperation of variables method motivates us to study fourier series.

Definition 4.1. The *trigonometric* polynomials of degree $2n$ on $[-\pi, \pi]$ are

$$\sum_{k=-n}^{n} c_k e^{ikx} \ x \in [-\pi, \pi]$$

The real version of a fourier series is

$$a_0 + \sum_{k=1}^{n} (a_k \cos(kx) + b_k \sin(kx))$$

Using De Moivre's formulas we can transition between one and the other, and this tells us that the complex fourier series is real valued iff $c_{-k} = \overline{c_k}$.

Essentially, fourier series theory is supercharged linear algebra. The inner product is the integral from $-\pi$ to π, and e^{ikx} are an orthogonal basis. This also tells us how to compute the coefficients if we wish to approximate a function by a fourier series. (To be more precise, we do this backwards. We assume that a function has a fourier series, then we show that the coefficients are given by, essentially, taking the inner product with the above 'basis').

$$\int_{-\pi}^{\pi} e^{ikx} e^{-ilx} dx = \begin{cases} 2\pi & k = l \\ 0 & k \neq l \end{cases}$$

This is orthogonality with respect to the inner product

$$(f, g) \mapsto \int_{-\pi}^{\pi} f(x)\overline{g(x)} dx$$

Let ϕ be a function that has a fourier series. The coefficients are then given by

$$\widehat{\phi}(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \phi(x) e^{-ikx} dx$$

We denote by $S_n(\phi)(x)$ the truncated fourier series:

$$S_n(\phi)(x) = \sum_{k=-n}^{n} \widehat{\phi}(k) e^{ikx}$$

It is easy to check that if ϕ is real-valued then indeed $\overline{\widehat{\phi}(k)} = \widehat{\phi}(-k)$.

The fourier coefficients are 'optimal' in the sense that they minimise the mean squared error.

Theorem 4.1. *Let $\phi \in C([-\pi, \pi], \mathbb{C})$. Then the fourier coefficients minimise*

$$\int_{-\pi}^{\pi} \left| \phi(x) - \sum_{k=-n}^{n} c_k e^{ikx} \right|^2 dx$$

among all possible choices of c_i.

The proof of this theorem is just algerbaic manipulation. However, as a consequence we get Bessel's Inequality:

$$2\pi \sum_{k \in \mathbb{Z}} |\widehat{\phi(k)}|^2 \leq \int_{-\pi}^{\pi} |\phi(x)|^2 dx$$

Equality holds above if

$$\lim_{n \to \infty} \int_{-\pi}^{\pi} |\phi(x) - S_n(\phi)(x)|^2 dx = 0$$

In other words, Bessel's Inequality is an equality (known as Parseval's Identity) if the Fourier Series of ϕ converges to ϕ in the mean-squared sense.

Another consequence of Bessel's Inequality is that the *Riemann-Lebesgue Lemma.* The fourier coefficients decay to zero:

$$\lim_{k \to \pm\infty} \widehat{\phi(k)} = 0$$

Theorem 4.2. *Let $\phi \in C^1(\mathbb{R})$ be 2π-periodic. Then $S_n(\phi)(x) \to \phi(x)$ for all $x \in [-\pi, \pi]$.*

The above theorem gives us conditions for pointwise convergence of fourier series. It is proved using the Dirichlet Kernel, which is omitted. Assuming stronger regularity conditions on ϕ, we can get uniform convergence of the fourier series.

Lemma 4.3. *Let $\phi \in C^s(\mathbb{R})$ be 2π-periodic. Then the coefficients decay as follows.*

$$|\widehat{\phi(k)}| \leq \frac{C}{|k|^s}, \quad \text{where } C = \left\| \phi^{(s)}(x) \right\|_\infty$$

Sketch of Proof. Use the definition of $\widehat{\phi(k)}$, then integrate by parts, noticing that the first part of the integral is 0 due to 2π-periodicity. □

Theorem 4.4. *Let $\phi \in C^2(\mathbb{R})$ be 2π-periodic. Then $S_n(\phi) \to \phi(x)$ uniformly on $[-\pi, \pi]$.*

Sketch of Proof. This follows from the above Lemma and applying definitions. □

Well and truly there are three tricks in this module: integration by parts, the Leibniz Rule, and seperating variables.

5. The Heat Equation

The heat equation is derived in the same way as the transport equation, simply with a different constitutive law - Fourier's Law.

Finding a solution for the boundary value problem with homogeneous dirichlet boundary conditions comes from seperating variables. In the case of inhomogeneous dirichlet boundary conditions, things are somewhat different, and this leads naturally to solving the inhomogeneous BVP with homogeneous initial and boundary conditions. We deal with using *Duhamel's Principle.*

We attempt to find a function $u : [0, L] \times [0, \infty) \to \mathbb{R}$ such that

$$\begin{aligned} &\partial_t u(x,t) = k\partial_{xx}u(x,t) + f(x,t), \quad (x,t) \in (0,L) \times (0,\infty) \\ &u(x,0) = 0 \\ &u(0,t) = u(L,t) = 0 \end{aligned} \tag{9}$$

We set up a modified family of problems, by making the inhomogenous part into an initial condition. For $\tau > 0$,

$$\begin{aligned} &\partial_t v(x,t;\tau) = k\partial_{xx}v(x,t;\tau), \quad (x,t) \in (0,L) \times (\tau,\infty) \\ &v(x,\tau;\tau) = f(x,\tau) \ x \in (0,L) \\ &v(0,t;\tau) = v(L,t;\tau) = 0 \end{aligned} \tag{10}$$

This is a problem that we can solve. We relate this back to the original problem with the following (frankly incredible) theorem:

Theorem 5.1 (Duhamel's Principle)**.** *Let $v(x,t;\tau)$ be a solution to* (10)*. Then*

$$u(x,t) = \int_0^t v(x,t;\tau)d\tau$$

solves (9)*.*

The Cauchy Problem

The derivation of the solution requires significantly more theory than was ever developed in this module. So, I just state everything. We are trying to find a function $u : \mathbb{R} \times (0, \infty) \to \mathbb{R}$ such that

$$\begin{aligned} &\partial_t u(x,t) = k\partial_{xx}u(x,t), \quad (x,t) \in \mathbb{R} \times (0,\infty) \\ &u(x,0) = \phi(x) \end{aligned}$$

Set

$$S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{\frac{-x^2}{4kt}}$$

The Cauchy problem is then solved by

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy$$

Energy Methods

The energy for the heat equation is given by

$$E_{HE}(t) = \frac{1}{2}\int_0^t (u(x,t))^2 dx \tag{11}$$

This is a decreasing function of t (take the derivative wrt t, then integrate by parts... again), and this fact can be used to show uniquess of solutions to the BVP.

The Maximum Principle for the Heat Equation

We return to the BVP with homogeneous dirichlet boundary conditions. Define the space-time rectangle $V_{L,T}$ as

$$V_{L,T} = (0,L) \times (0,T]$$

We define the parabolic boundary, $\Gamma_{L,T} = \overline{V_{L,T}} \setminus V_{L,T}$. The maximum principle says that the maximum of any solution to the heat equation is obtained on the parabolic boundary. We can use this to show uniqueness, and that the heat equation is well-posed (gives us a bound on the difference between two solutions in terms of their boundary and initial conditions - this is exactly the definition of 'stability' found in well-posedness).

Radially Symmetric Problems

There are two (really one) types of equations we consider here. The Laplace equation and the Poisson Equation. Write in terms of polar co-ordinates. Maximum principle, seperation of variables. Done.

6. Appendix: Results from other Modules

Theorem 6.1 (The Fundamental Theorem of Calculus, Part 1)**.** *This relates the integrand to its antiderivative. Let $f : [a,b] \to \mathbb{R}$ be integrable. Define a function F as follows:*

$$F(x) = \int_a^x f(t)dt$$

Then F is continuous, and if f is continuous at $u \in (a,b)$, then F is differentiable at u with $F'(u) = f(u)$.

As a consequence, we have the following. Let $g : I \times J \to \mathbb{R}$ be a continuous function.

$$\frac{d}{da}\int_a^b g(x,y)dy = -g(x,a) \qquad\qquad \frac{d}{db}\int_a^b g(x,y)dy = g(x,b)$$

Theorem 6.2 (The Fundamental Theorem of Calculus, Part 2)**.** *This relates the derivative to the integral. Let $F : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$, differentiable on (a,b) such that $F' = f$. Then if f is integrable, we have*

$$\int_a^b f(t)dt = F(b) - F(a)$$

Theorem 6.3 (Leibniz' Integral Rule)**.** *Let $I, J \subset \mathbb{R}$ be intervals. Let $a : I \to \mathbb{R}$, $b : J \to \mathbb{R}$ be differentiable, let $g : I \times J \to \mathbb{R}$ be continuous, such that $\partial_y g$ is continuous. Then*

$$\frac{d}{dx} \int_{a(x)}^{b(x)} g(x,y)dy = g(x, b(x))b'(x) - g(x, a(x))a'(x) + \int_{a(x)}^{b(x)} \partial_x g(x,y)dy$$

Before the proof, a remarks: firstly, in the case where a, b are constant functions, this tells us that we can 'pull in' the derivative.

$$\frac{d}{dx} \int_a^b g(x,y)dy = \int_a^b \partial_x g(x,y)dy$$

Proof. We interpret this as an application of the multivariable chain rule, and apply the FTC (pt 1). $\square$